JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in **E-mail:** info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn, A Chance to Grow"
Session: 2023-27

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

B. Tech.

In

Computer Science and Engineering

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VI Semester Computer Science & Engineering

Sr. No.	Category of Subject	Course Code	Course Name	Teaching Scheme			Evaluation Scheme				Credit
				L	T	P	CA	MSE	ESE	Total	
1	PCC	CS6T001	Cloud Computing	2	0	0	20	20	60	100	2
2	PCC	CS6T002	Software Testing & Quality Assurance	2	0	0	20	20	60	100	2
3	PCC	CS6T003	AI & Applications	2	0	0	20	20	60	100	2
4	PEC	CS6E003	PEC-III	3	0	0	20	20	60	100	3
5	PEC	CS6E004	PEC-IV	3	0	0	20	20	60	100	3
6	PEC	CS6E005	PEC-V	2	0	0	20	20	60	100	2
7	MDM	CS6M005	Supply Chain & Inventory Optimization	2	0	0	20	20	60	100	2
8	MDM	CS6M006	Project Quality & Control	2	0	0	20	20	60	100	2
9	PCC	CS6L004	Cloud Computing Lab	0	0	2	60	-	40	100	1
10	PCC	CS6L005	NoSQL Databases Lab	0	0	2	60	-	40	100	1
11	VSEC	CS6L006	Full Stack Development	0	0	4	60	-	40	100	2
				18	**0**	**8**	**340**	**160**	**600**	**1100**	**22**

PEC	Subject Code	Subject Name
III	CS6E003A	Middleware Technology
	CS6E003B	Digital Image Processing
	CS6E003C	Soft Computing
	CS6E003D	Deep Learning
IV	CS6E004A	No SQL & New SQL
	CS6E004B	Semantic Web
	CS6E004C	Parallel Computing
	CS6E004D	Robotic Automation System
V	CS6E005A	Data Warehousing & Mining
	CS6E005B	Augmented Reality
	CS6E005C	Grid Computing
	CS6E005D	Compiler Design

Secretary, BOS,
CSE Dept.
JDCOEM, Nagpur

Chairman, BOS,
CSE Dept.
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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6T001**	**Cloud Computing**	2	0	0	**2**

Prerequisites for the course	
1.	Basics of Cloud computing platforms, such as Amazon Web Services, Microsoft Azure, and Google Cloud..
2.	Understand how to host network devices like virtual firewalls, routers, and bandwidth from the cloud

Prior Reading Material/useful links	
1.	https://www.mygreatlearning.com/cloud-computing/courses
2.	https://onlinecourses.nptel.ac.in/noc21_cs14/preview

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understanding the Fundamentals of Cloud Computing.
2	CO2	Student will be able to Analysing Cloud Services and Virtualization Technology.
3	CO3	Student will be able to Utilizing Cloud Platforms for Application Deployment.
4	CO4	Student will be able to Managing Cloud Migration and Security.
5	CO5	Student will be able to Exploring Cloud Computing Tools and Future Trends.

Syllabus:

Course Contents		Hours
Unit I	**Defining Cloud Computing:** Cloud computing in a nutshell, Cloud type - NIST Model, cloud cube model, deployment model, service model, Characteristics of cloud computing, cloud computing stack, open stack	**6**

Unit II	**Understanding Services and Virtualization Technology:** Understanding services and applications, defining Infrastructure as a Service (IaaS),Platform as a service, Software as a Service, Identity as a Service, Compliance as a Service, Using virtualization technologies, Load balancing and virtualization, understanding Hypervisors, Salesforce.com versus Force.com, SaaS versus PaaS.	**7**
Unit III	**Using Cloud Platform:** Using Google web services, using Amazon web services, using Microsoft cloud services, Aneka integration of private and public cloud.	**3**
Unit IV	**Cloud Migration:** Broad approaches to migration, seven steps model of migration, Applications in the cloud, Application in cloud API, Cloud Security and Storage, Securing the cloud, securing data, working with cloud based storage.	**4**
Unit V	**Cloud Computing Tools and Future Cloud:** Open source cloud computing platform - Eucalyptus, Open Nebula, Programming in the cloud Map Reduce Dryad. Future cloud - Future trends in cloud computing, defining the mobile market, using Smart phones with the cloud.	**4**

Text Books	
1.	"Cloud Computing Bible", Barrie Sosinsky; Wiley India Pvt. Ltd
2.	"Cloud Computing - Principals and Paradigms", Rajkumar Buyya, James Broberg, Andrzej Goscinski; Wiley India Pvt. Ltd.
3.	Cloud Computing, A Hands on Approach, Bahga, Madisetti, University Press

Reference Books	
1.	"Mastering Cloud Computing", Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Tata McGraw Hill
2.	"Cloud Computing - A practical approach for learning and implementation", A. Shrinivasan, J. Suresh; Pearson
3.	Cloud Computing - Fundamentals, Industry approach and trends", Rishabh Sharma; Wiley India Pvt. Ltd.

Useful Links	
1.	https://www.geeksforgeeks.org/cloud-computing/
2.	https://www.techtarget.com/searchcloudcomputing/definition/cloud-computing
3.	https://www.tutorialspoint.com/cloud_computing/cloud_computing_tutorial.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6T002**	**Software Testing & Quality Assurance**	2	0	0	**2**

Prerequisites for the course	
1.	Basic understanding of software engineering principles.
2.	Basic experience with some test automation tools.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/software-testing-vs-quality-assurance/
2.	https://www.tutorialspoint.com/software_testing/software_testing_qa_qc_testing.htm

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understanding the Basics of Software Testing and Its Importance.
2	CO2	Student will be able to gain a thorough understanding of unit testing and Debugging Techniques.
3	CO3	Student will be able to Applying System Integration Testing and Test Design Techniques.
4	CO4	Students will learn to categorize and perform various system tests.
5	CO5	Student will be able to Performing Acceptance Testing and Ensuring Software Quality.

Syllabus:

Course Contents		Hours
Unit I	**Basic concepts of Testing:** Need of Testing, Basic concepts- errors, faults, defects, failures, objective of testing, central issue in testing, Testing activities, V-Model, Monitoring and Measuring Test Execution, Test tools and Automation, Limitation of Testing.	**5**
Unit II	**Unit Testing:** Concepts of Unit Testing, Static Unit Testing, Defect Prevention, Dynamic Unit Testing, Mutation Testing, Debugging, Tools for Unit Testing.	**3**

Unit III	**Control Flow and Data Flow Testing:** Control Flow Testing, Control Flow Graphs, Path in Control Flow Graph, Path Selection Criteria, Generating Test Input and Data Selection Examples **Introduction to Data Flow Testing**, Data Flow Graphs, Data Flow Testing Criteria, Comparison of Data Flow Test Selection Criteria.	**5**
Unit IV	**System Test Categories and Test Design:** Taxonomy of system test, Basic Test, Functionality test, Robustness test, Performance test, Scalability test, Stress test, Load and Stability test, Reliability test, Regression test, Documentation Test. Test Design: Test cases, Necessity of test case documentation, Test case design methods, Functional specification based test case design, Use case bases, Application based test case design, Level of test execution.	**6**
Unit V	**Acceptance Testing and Software Quality:** Types of acceptance testing, Acceptance criteria, Acceptance test plan and execution, Special Tests: Client server testing, Web application testing and Mobile application testing, fire view of software quality, ISO-9126 quality characteristics, ISO-9000:2000 software quality standard, ISO - 9000:2000 fundamentals.	**5**

Text Books	
1.	Software Testing and Quality Assurance by Kshirsager Naik and Priyadarshini Tripathi (Wiley)
2.	Software Testing Concepts and Tools by Nageswara Rao Pusuluri (Dream Tech Press).
3.	Software Testing Principles, Techniques and tools, 1st Edition, by M. G. Limaye McGraw Hills.

Reference Books	
1.	"Foundations of Software Testing" 2E by Aditya P. Mathur , Pearson Education.
2.	Effective Methods for Software Testing- William E Perry, (Wiley). 2. Software Testing Tools by Dr. K. V. K. K. Prasad (Dream Tech)

Useful Links	
1.	https://www.intertek.com/software/quality-assurance/
2.	https://qalified.com/blog/what-is-qa-testing/
3.	https://testsigma.com/blog/software-testing-and-quality-assurance/

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6T003**	**AI & Applications**	2	0	0	**2**

Prerequisites for the course	
1.	Knowledge of AI Tools and Frameworks.
2.	Familiarity with big data frameworks.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/applications-of-ai/
2.	https://esdst.eu/arti%EF%AC%81cial-intelligence-the-future-and-its-applications/

Course Outcomes:

Sr. No	Course Outcome No.	CO statement
1	CO1	Student will be able to Understand the AI and AI Problem.
2	CO2	Student will be able to Analyse the data using predicate logic knowledge.
3	CO3	Student will be able to Solve the problem using Bayes and DST Probabilistic Reasoning
4	CO4	Students will learn to Apply Natural Language Processing kit on given sentence.
5	CO5	Student will be able to Recall and understand the concept of Expert System.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Artificial Intelligence:** Definition of AI, History of AI, examples of AI problems. Current trends in Artificial Intelligence, Intelligent Agents, types of agents. Problem solving by search: Uninformed Search: Depth First Search (DFS), Breadth First Search (BFS), Informed Search: Best First Search, A*. Local Search: Hill Climbing, Problem Reduction Search: AO*, Population Based Search: Adversarial Search: Game Playing-Min Max Algorithm, Alpha-Beta Pruning.	**7**
Unit II	**Knowledge Representation:** Types of Knowledge, Knowledge Representation Techniques, Propositional Logic, syntax, inference, Predicate Logic, Semantic nets, Frames, Knowledge representation issues, Rule based systems.	**7**

Unit III	**Reasoning Under Uncertainty:** Basics of Probability Theory, Probabilistic Reasoning: Bayes Rules, Probabilistic Reasoning: Bayesian statistic, Dempster-Shafer Theory, Planning: Spare, Block world, Planning with state space search, Representation of Planning, Partial-order Planning.	**4**
Unit IV	**Planning and Learning:** Learning: Introduction to Learning, Types of Learning, Rote Learning, Symbol Based Learning, Identification Trees, Explanation Based Learning, Transformational Analogy.	**3**
Unit V	**Applications :** Natural Language Processing, Language Models, Text classification, Information Retrieval, information extraction, Expert System: Introduction, Phases in Building Expert Systems.	**3**

Text Books	
1.	Artificial Intelligence: A Modern Approach Stuart R. &Peter Norvig 2nd Edition. Pearson Education
2.	Artificial Intelligence Elaine Rich, Kevin Knight, Shivshankar B Nair, 3rd Edition. McGraw Hill,
3.	Artificial Intelligence Elaine Rich, Kevin Knight, 2nd Edition. Tata McGraw Hill

Reference Books	
1.	Artificial Intelligence Patrick H. Winston 3rd Edition Pearson Education.
2.	A First Course in Artificial Intelligence Deepak Khemani McGraw Hill Publication

Useful Links	
1.	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
2.	https://www.coursera.org/learn/introduction-to-ai
3.	https://mrcet.com/downloads/digital_notes/IT/(R17A1204)%20Artificial%20Intelligence.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6E003A**	**Elective-III (Middleware Technology)**	3	0	0	**3**

Prerequisites for the course	
1.	Basics of Java Programming
2.	Fundamental of client server computing model
3.	Analysis of RMI and JDBC Connectivity.

Prior Reading Material/useful links	
1.	https://www.ibm.com/in-en/topics/middleware
2.	https://www.talend.com/resources/what-is-middleware/
3.	https://www.talend.com/resources/what-is-middleware/

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students are able to Choose appropriate client server computing model for given problem.
2	CO2	Students are able to Design a dynamic remote application with RMI and JDBC Connectivity.
3	CO3	Students are able to Develop client server applications using C#.net
4	CO4	Students are able to Select appropriate language for homogeneous and heterogeneous objects.
5	CO5	Students are able to Develop real time projects by combining CORBA and database interfacing

Syllabus:

Course Contents		Hours
Unit I	**Introduction to client server computing**: Evolution of corporate computing models from centralized to Distributed computing, client server models. Benefits of client server computing , pitfalls of client server Programming. Advanced Java: Review of Java concept like RMI , and JDBC.	**8**

Unit II	**Introducing C# and the .NET Platform:** Understanding .NET Assemblies, Object –Oriented Programming with C#, Callback Interfaces. Building c# applications: Type Reflection, Late Binding, and Data Access with ADO.NET .	**8**
Unit III	**Core CORBA / Java:** Two types of Client/ Server invocations-static , dynamic. The static CORBA, first CORBA program, ORBlets with Applets, Dynamic CORBA-The portable count, the dynamic count Existential CORBA: CORBA initialization protocol, CORBA activation services, Introduction to SOA	**8**
Unit IV	**Java Bean Component Model:** Events, properties, persistency, Introspection of beans, CORBA Beans. CORBA: Object transaction monitors CORBA OTM's, CORBA OTM's.	**3**
Unit V	**EJB Java Bean**: Component Model, EJB Architecture, Session Bean, Java Message Service, Message Driven Bean, Entity Bean	**3**

Text Books	
1.	Client/Server programming with Java and CORBA Robert Orfali and Dan Harkey, John Wiley & Sons , SPD 2nd Edition .
2.	Java programming with CORBA 3rd Edition, G.Brose, A Vogel and K.Duddy, Wiley-dreamtech, India John Wiley and sons.

Reference Books	
1.	Distributed Computing, Principles and applications, M.L.Liu, Pearson Education.
2.	Client/Server Survival Guide 3rd edition Robert Orfali Dan Harkey & Jeri Edwards, John Wiley & Sons 3. C# Preciesely Peter Sestoft and Henrik I . Hansen , Prentice Hall of India.

Useful Links	
1.	https://books.google.com/books?id=wsnBA4aD2h0C&printsec=frontcover&dq=Middleware+Technologies&hl=en&newbks=1&newbks_redir=1&sa=X&ved=2ahUKEwjeyZ2nmbj-AhXypOkKHUgrDqkQ6AF6BAgHEAI
2.	https://www.google.co.in/books/edition/The_Complete_Book_of_Middleware/Gc886KgsdcsC?hl=en&gbpv=1&dq=Middleware+Technologies&printsec=frontcover
3.	https://www.google.co.in/books/edition/Introduction_to_Middleware/AZgnDwAAQBAJ?hl=en&gbpv=1&dq=Middleware+Technologies&printsec=frontcover
4.	https://www.google.co.in/books/edition/Middleware_for_Communications/bh8_0FD3qgQC?hl=en&gbpv=1&dq=Middleware+Technologies&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6E003B**	**Elective-III (Digital Image Processing)**	3	0	0	**3**

Prerequisites for the course	
1.	Study the fundamental concepts of a digital image processing system.
2.	Categorize various compression techniques.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/digital-image-processing-basics/
2.	https://www.tutorialspoint.com/dip/index.htm
3.	https://www.javatpoint.com/digital-image-processing-tutorial

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Student will able to understand the fundamental concepts of a digital image processing system.
2	CO2	Student will able to Analyse images in the frequency domain using various transforms
3	CO3	Student will able to Evaluate the techniques for image enhancement and image restoration.
4	CO4	Student will able to Categorize various compression techniques.
5	CO5	Student will able to Interpret Image compression standards.

Syllabus:

	Course Contents	Hours
Unit I	**Introduction to Digital Image Processing:** Fundamentals of Elements of Digital Image, Image As Data, Pixels, Components Of Digital Image, Types Of Image Representation, Measures Of Image, Neighbors of pixel adjacency connectivity, regions and boundaries, Distance measures, Application Of Digital Image Processing.	**8**
Unit II	**Matlab Basics:** Introduction to Data Types, Operators, Matrices, File, I/O, Image Processing Toolbox	**5**

Unit III	**Image Enhancement Techniques:** In spatial domain: Basic gray level transformations, Histogram processing, using arithmetic/Logic operations, smoothing spatial filters, Sharpening spatial filters. In Frequency domain: Introduction to the Fourier transform and frequency domain concepts, smoothing frequency-domain filters, Sharpening frequency domain filters.	**8**
Unit IV	**Image Filtering Techniques:** Low Pass Filters – Smoothing, High Pass Filters - Edge Detection, Sharpening; Image Restoration: Noise Models, Model of Image Degradation/Restoration Process, Noise Reduction, Inverse Filtering, M Minimum Mean Square Error (Weiner) Filtering.	**8**
Unit V	**Colour Image processing:** Colour fundamentals, Colour models, Representation of Color in Images, Colour transformation, Smoothing and Sharpening, Colour segmentation. Image Morphology: Different Morphological Algorithm, Morphological Measures	**7**

Text Books	
1.	Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing, Pearson Education, Third Edition, 2008.
2.	Anil K. Jain, Fundamentals of Digital Image Processing', Pearson 2002. Gonzalez & Woods - Digital Image Processing Using Matlab
3.	Bhabatosh Chanda and Dwijesh Majumder - Digital Image Processing
4.	Kenneth R. Castleman, Digital Image Processing, Pearson, 2006

Reference Books	
1.	Rafael C. Gonzalez, Richard E. Woods, Steven Eddins,' Digital Image Processing using MATLAB', Pearson Education, Inc., 2004.
2.	William K. Pratt, Digital Image Processing' , John Wiley, New York, 2002
3.	D. E. Dudgeon and RM. Mersereau, Multidimensional Digital Signal Processing', Prentice Hall Professional Technical Reference, 1990.

Useful Links	
1.	https://www.bharathuniv.ac.in/colleges1/downloads/courseware_ece/course_outcome/core_electives_3/BEC007%20CO%20DIGITAL%20IMAGE%20PROCESSING.pdf
2.	https://www.cis.rit.edu/class/simg712-90/notes/12-Basic_Principles_DIP.pdf
3.	https://www.google.co.in/books/edition/Image_Processing/smBw4xvfrIC?hl=en&gbpv=1&dq=What+is+the+basic+principle+of+digital+image+processing%3F&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6E003C**	**Elective-III (Soft Computing)**	3	0	0	**3**

Prerequisites for the course	
1.	Proficiency with algorithms.
2.	Programming skills in C, C++, or Java, MATLAB, etc.

Prior Reading Material/useful links	
1	https://cse.iitkgp.ac.in/~dsamanta/courses/sca/resources/slides/01%20Intro.pdf
2	https://cse.iitkgp.ac.in/~dsamanta/courses/sca/resources/slides/02%20FL%20Introduction.pdf
3	https://cse.iitkgp.ac.in/~dsamanta/courses/sca/resources/slides/06%20ANN%20Introduction.pdf

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Student will able to Understanding Soft Computing and its Key Characteristics.
2	CO2	Student will able to Mastering Fuzzy Logic Concepts and Applications.
3	CO3	Student will able to Implementing and Applying Genetic Algorithms.
4	CO4	Student will able to Solving Multi-Objective Optimization Problems Using Evolutionary Algorithms.
5	CO5	Student will able to Exploring Artificial Neural Networks and their Applications.

Syllabus:

	Course Contents	Hours
Unit I	**Introduction to Soft Computing:** Define Soft Computing, Characteristics of Soft computing**,** Example, Applications of soft computing, Need of soft computing, Elements of soft computing, Soft computing vs hard computing.	**5**

Unit II	**Fuzzy logic:** Introduction to Fuzzy logic, Fuzzy sets and membership functions, Operations on Fuzzy sets, Fuzzy relations, rules, propositions, implications and inferences, Defuzzification techniques, Fuzzy logic controller design, Some applications of Fuzzy logic.	**8**
Unit III	**Genetic Algorithms:** Concept of "Genetics" and "Evolution" and its application to proablistic search techniques, Basic GA framework and different GA architectures, GA operators: Encoding, Crossover, Selection, Mutation, etc, Solving single-objective optimization problems using GAs.	**8**
Unit IV	**Multi-objective Optimization Problem Solving:** Concept of multi-objective optimization problems (MOOPs) and issues of solving them. Multi-Objective Evolutionary Algorithm (MOEA), Non-Pareto approaches to solve MOOPs, Pareto-based approaches to solve MOOPs, Some applications with MOEAs.	**8**
Unit V	**Artificial Neural Networks:** Biological neurons and its working, Simulation of biolgical neurons to problem soloving, Different ANNs architectures, Trainging techniques for ANNs, Applications of ANNs to solve some real life problems.	**7**

Text Books	
1.	Soft Computing, D. K. Pratihar, Narosa, 2008.
2.	Neural Networks, Fuzzy Logis and Genetic Algorithms : Synthesis, and Applications, S. Rajasekaran, and G. A. Vijayalakshmi Pai, Prentice Hall of India, 2007.
3.	Neural Networks and Learning Machines, (3rd Edn.), Simon Haykin, PHI Learning, 2011.
4.	An Introduction to Genetic Algorithms, Melanie Mitchell, MIT Press, 2000.

Reference Books	
1.	Fuzzy Logic: A Pratical approach, F. Martin, , Mc neill, and Ellen Thro, AP Professional, 2000.
2.	Foundations of Neural Networks, Fuzzy Systems, and Knowldge Engineering, Nikola K. Kasabov, MIT Press, 1998.
3.	Practical Genetic Algorithms, Randy L. Haupt and sue Ellen Haupt, John Willey & Sons, 2002.

Useful Links	
1.	https://www.geeksforgeeks.org/need-for-soft-computing/
2.	https://www.elprocus.com/soft-computing/
3.	https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://ggu.ac.in/gguold/download/Class-Note14/Soft%2520Computing-

	%2520Dr.%2520H.s.%2520Hota%252028.08.14.pdf&ved=2ahUKEwi2rsmY7_eLAxUmT2cHHbqXKy4QFnoECDEQAQ&usg=AOvVaw1jbPBtSTqkgpQyJnqrwO9o

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6E003D**	**Elective-III (Deep Learning)**	3	0	0	**3**

Prerequisites for the course	
1.	Basic knowledge of how neural networks work.
2.	Skills in cleaning and preparing data for models.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/introduction-deep-learning/
2.	http://neuralnetworksanddeeplearning.com/
3.	https://www.javatpoint.com/deep-learning

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will able to Gain basic knowledge of Neural Networks.
2	CO2	Students will Apply appropriate deep learning algorithms for analysing the data for a variety of problems.
3	CO3	Students will be able to Use the unsupervised deep learning models and analyse the performance.
4	CO4	Students will be able to Gain familiarity with recent trends and applications of Deep Learning.
5	CO5	Students will be Implement deep learning algorithms, understand neural networks and traverse the layers of data abstraction.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Deep Learning:** Evolution of AI, Machine Learning vs Deep Learning, Deep Learning types, History of Deep Learning, Model of Artificial	**8**

	Neuron, Activation functions, McCulloch Pitts Neuron, Perceptrons, Forward and Back Propagation, Gradient Descent (GD).	
Unit II	**Deep Learning Models:** Introduction to CNNs, CNN operation,s Architectures, Simple Convolution Network, Deep Convolutional Models – LeNet, ResNet, AlexNet and others. Introduction to RNNs, Back propagation through time (BPTT), Vanishing and Exploding Gradients, Gated recurrent Unit, Long Short-Term Memory Networks, Dropouts.	**8**
Unit III	**Deep Unsupervised Learning:** Autoencoders (standard, sparse, denoising, contractive, etc), Motivation for Sampling, Markov Networks, Variational Autoencoders. Introduction of GANs(Generative Modeling) , Different Types of GANs, Components of GANs, Training and Prediction of GANs, Brief on GAN Loss Function, Challenges Faced by GANs, Application of GANs.	**8**
Unit IV	**Regularization for Deep Learning:** Overview of Overfitting, Types of biases, Bias Variance Tradeoff Regularization Methods: L1, L2 regularization, Parameter sharing, Dropout, Weight Decay, Batch normalization, Early stopping, Data Augmentation, Adding noise to input and output.	**6**
Unit V	**Applications of Deep Learning:** Image segmentation, object detection, Automatic Image Captioning , Capture and Representation, Image Processing, Introduction to NLP, Word Vector representation ,Speech recognition, Video Analytics.	**6**

Text Books	
1.	Goodfellow, I., Bengio,Y., and Courville, A., Deep Learning, MIT Press, 2016.
2.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
3.	Deng & Yu, Deep Learning: Methods and Applications, Now Publishers, 2013.

Reference Books	
1.	Michael Nielsen, Neural Networks and Deep Learning, Determination Press, 2015.
2.	Neural Networks: A Systematic Introduction, Raúl Rojas, 1996
3.	Pattern Recognition and Machine Learning, Christopher Bishop, 2007

Useful Links	
1.	https://deepai.org/machine-learning-glossary-and-terms/deep-learning
2.	https://journalofbigdata.springeropen.com/articles/10.1186/s40537-021-00444-8

VISION	MISSION
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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6E004B**	**Elective-IV (Semantic Web)**	3	0	0	**3**

Prerequisites for the course	
1.	Study fundamentals of Semantic Web technologies.
2.	Creating structured web documents in XML

Prior Reading Material/useful links	
1.	https://www.techtarget.com/searchcio/definition/Semantic-Web
2.	https://www.analyticssteps.com/blogs/what-semantic-web-working-importance-and applications
3.	https://www.techtarget.com/searchcio/definition/Semantic-Web

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Student will able to Understand the fundamentals of Semantic web
2	CO2	Student will able to Creating structured web documents in XML
3	CO3	Student will able to Apply ontology engineering to various problems.
4	CO4	Student will able to Understand Semantic Web query languages (SPARQL)
5	CO5	Student will able to Program semantic applications with Java and Jena API.

Syllabus:

Course Contents		Hours
Unit I	**Semantic Web Vision**: Todays' web, Examples of semantic web from today's web, Semantic web technologies, layered approach Structured web documents in XML: The XML language, Structuring, Namespaces, Querying and Addressing XML documents, Processing	**7**
Unit II	**Describing Web Resources:** Introduction, RDF: Basic Ideas,RDF: XML-Based Syntax,RDF serialization, RDF Schema: Basic Ideas,RDF Schema: The Language ,RDF and RDF Schema	**7**

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Unit III	**Logic and Inference Rules:** Introduction, Monotonic Rules syntax, semantics & examples, Nonmonotonic rules – syntax & examples, Encoding in XML.	**7**
Unit IV	**Ontology Engineering**: Introduction, Manual construction of Ontology, Reusing existing ontology, using Semi-automatic methods, Knowledge semantic web architecture	**7**
Unit V	**SPARQL, Ontology Language**: SPARQL simple Graph Patterns, Complex Graph Patterns, Group Patterns, Queries with Data Values, Filters OWL Formal Semantics. SchemaWeb Ontology Language: Introduction, OWL language, Examples, OWL in OWL, Future extensions.	**8**

Text Books	
1.	A Semantic web Primer: Grigoris Antoniou and Frank Van Hermelen , MIT Press
2.	Semantic Web programming, John Hebler et.el, Wiley
3.	Foundations of Semantic Web Technologies, Pascal Hitzler, Markus Krötzsch, Sebastian Rudolph, CRC Press
4.	Michael C. Daconta, Leo J. Obrst, and Kevin T. Smith, "The Semantic Web: A Guide to the Futureof XML, Web Services, and Knowledge Management", Fourth Edition, Wiley Publishing, 2003.

Reference Books	
1.	John Davies, Rudi Studer, and Paul Warren John, "Semantic Web Technologies: Trends and Research in Ontology-based Systems", Wiley and Son's, 2006.
2.	John Davies, Dieter Fensel and Frank Van Harmelen, "Towards the Semantic Web: Ontology- Driven Knowledge Management", John Wiley and Sons, 2003.
3.	Semantic Web and education By Vladan Devedzic · 2006
4.	Semantic Web Technologies for E-learning, Darina Dicheva, Jim E. Greer, Riichiro Mizoguchi, IOS Press

Useful Links	
1.	https://mrcet.com/downloads/digital_notes/CSE/IV%20Year/SEMANTIC%20WEB&SOCIAL%20NETWORKS.pdf
2.	https://www.google.co.in/books/edition/Semantic_Web_Engineering_in_the_Knowledg/ipN3GWuQXgC?hl=en&gbpv=1&dq=semantic+web+course+objectives&printsec=frontcover
3.	https://www.google.co.in/books/edition/Semantic_Web_Services/wglTWruS3UC?hl=en&gbpv=1&dq=semantic+web+course+objectives&printsec=frontcover

4.	https://www.google.co.in/books/edition/Social_Networks_and_the_Semantic_Web/giOwMTn2PNYC?hl=en&gbpv=1&dq=semantic+web+course+objectives&printsec=frontcover

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6E005B**	**Elective-V (Augmented Reality)**	2	0	0	**2**

Prerequisites for the course	
1.	Basic knowledge & Information of virtual reality.

Prior Reading Material/useful links	
1.	https://www.techtarget.com/whatis/definition/augmented-reality-AR
2.	https://arvr.google.com/ar/
3.	https://www.ptc.com/en/technologies/augmented-reality

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Student will able to understand the basic concept and framework of virtual reality
2	CO2	Student will able to categories the technology for multimodal user interaction and perception in VR.
3	CO3	Student will able to evaluate algorithmic strategies to solve a given problem.
4	CO4	Student will able to apply VR Tools in real time environment.
5	CO5	Student will able to implement application of AR/VR technology with hands on experience through more informative and practical exploration.

Syllabus:

	Course Contents	Hours
Unit I	**Introduction**: VR and AR Fundamentals, Differences between AR/VR Selection of technology AR or VR AR/VR characteristics Hardware and Software for AR/VR introduction. Requirements for VR/AR. Benefits and Applications of AR/VR. AR and VR case study.	**08**

Unit II	**Fundamentals of Computer Graphics:** Real time rendering technology; Principles of Stereoscopic Display; Software and Hardware Technology on Stereoscopic Display.	**06**
Unit III	**Software Technologies:** Database, World Space, World Coordinate, World Environment, Objects, Geometry, Position / Orientation, Hierarchy, Bounding Volume, Scripts and other attributes, VR Environment, VR Database, Tessellated Data, LODs, Cullers and Occluders, Lights and Cameras, Scripts, Interaction, Simple, Feedback, Graphical User Interface, Control Panel, 2D Controls, Hardware Controls, Room / Stage / Area Descriptions, World Authoring and Playback, VR toolkits, Available software in the market (Unity and Vuforia based) - Case Studies in AR, VR – Industrial applications, medial AR/VR, education and AR/VR.	**08**
Unit IV	**Geometric Modeling; Behavior Simulation;** Physically Based Simulationn Haptic & Force Interaction in Virtual Reality **Concept of haptic interaction:** Principles of touch feedback and force feedback; Typical structure and principles of touch/force feedback facilities in applications	**08**
Unit V	**VR Development Tools:** Frameworks of Software Development Tools in VR; Modeling Tools for VR; X3D Standard; Vega, Multi Gen	**06**

Text Books	
1.	Burdea, G. C. and P. Coffet. Virtual Reality Technology, Second Edition. Wiley-IEEE Press, 2003/2006.
2.	Alan B Craig, William R Sherman and Jeffrey D Will, Developing Virtual Reality Applications: Foundations of Effective Design, Morgan Kaufmann, 2009.
3.	Gerard Jounghyun Kim, Designing Virtual Systems: The Structured Approach, 2005.

Reference Books	
1.	Doug A Bowman, Ernest Kuijff, Joseph J LaViola, Jr and Ivan Poupyrev, 3D User Interfaces, Theory and Practice, Addison Wesley, USA, 2005.
2.	Oliver Bimber and Ramesh Raskar, Spatial Augmented Reality: Meging Real and Virtual Worlds, 2005.
3.	Burdea, Grigore C and Philippe Coiffet, Virtual Reality Technology, Wiley Interscience, India

Useful Links	
1.	https://mobidev.biz/blog/augmented-reality-development-guide

2.	https://firsthand.co/professions/augmented-reality-developers/requirements
3.	https://www.coursera.org/learn/augmented-reality
4.	https://www.sciencedirect.com/science/article/pii/S221282711830163X

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
VI	**CS6L004**	**Cloud Computing Lab**	0	0	2	**1**

Prerequisites for the course	
1.	Understand Basics of Discrete Mathematics
2.	Fundamental of SQL.

Prior Reading Material/useful links	
1.	https://shanmugha.edu.in/pdf/dept/cse/LAB_MANUALS/Cloudcomputinglabmanual_cse.pdf
2.	https://www.stannescet.ac.in/cms/staff/qbank/CSE/Lab_Manual/CS8711 CLOUD%20COMPUTING%20LABORATORY-850713797 CC%20LAB%20MANUAL%20(2).pdf

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Student will be able to Configure various virtualization tools such as Virtual Box, VMware workstation.
2	CO2	Student will be able to Design and deploy a web application in a PaaS environment.
3	CO3	Student will be able to Learn how to simulate a cloud environment to implement new schedulers.
4	CO4	Student will be able to Install and use a generic cloud environment that can be used as a private cloud.
5	CO5	Student will be able to Manipulate large data sets in a parallel environment..

Course Contents List of Practical's	Hours

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0	Introduction to Cloud Computing Lab.	**2**
1	Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows7 or 8.	**2**
2	Install a C compiler in the virtual machine created using virtual box and execute Simple Programs..	**2**
3	Install Google App Engine. Create hello world app and other simple web applications using python/java.	**2**
4	Register with Amazon AWS and create a windows/linux instance and connect with RDP and create S3 buckets	**2**
5	Use GAE launcher to launch the web applications..	**2**
6	Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim	**2**
7	Find a procedure to transfer the files from one virtual machine to another virtual machine.	**2**
8	Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version.	**2**
9	Installation and Configuration of Just cloud.	**2**
10	Install Hadoop single node cluster and run simple applications like wordcount.	**2**

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Cloud Computing | Thomas Erl, Zaigham Mahmood, Ricardo Puttini | Pearson
2.	Cloud Computing: Concepts, Technology, Security & Architecture, 2nd Edition - Pearson

Reference Books	
1.	Cloud Computing Computerized Notes for Computer Science & Engineering by Riyaz ·
2.	Cloud Computing Networking: Theory, Practice, and Development by Lee Chao · 2015

Useful Links

1.	https://srmvalliammai.ac.in/wp-content/uploads/2022/05/it8711-foss-and-cloud computing-lab-manual.pdf
2.	https://files.mlrit.ac.in/curriculum/mtech/CSE-MLR18/1-1/CLOUD-COMPUTING-LAB.pdf

Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
V	**CS6L005**	**NoSQL Databases Lab**	0	0	2	**1**

Prerequisites for the course	
1.	Basics Knowledge of Computer programming.
2.	Introduction to Computer programming Lab.

Prior Reading Material/useful links	
1.	https://www.javatpoint.com/machine-learning
2.	https://www.tutorialspoint.com/machine_learning/index.htm
3.	https://onlinecourses.nptel.ac.in/noc23_cs18/preview

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will be able to choose the concepts of computational intelligence in Machine Learning for Designing.
2	CO2	Students will be able to classify the data based on Bayes Theorem and other techniques.
3	CO3	Students will be able to apply ANN Techniques to address the real time problems.
4	CO4	Students will be able to build new software for Real time problems using Reinforcement Learning.
5	CO5	Students will be able to design different Board positions and different applications by using Analytical Learning.

List of Practical's

Course Contents		Hours
0	Introduction of NoSQL Databases Lab.	2

1	Installation of NoSQL Databases	2
2	Demonstrate various data pre-processing techniques for a given dataset	2
3	Implement Dimensionality reduction using Principle Component Analysis (PCA) method.	2
4	Write a Python program to demonstrate various Data Visualization Techniques.	2
5	Implement Simple and Multiple Linear Regression Models.	2
6	Develop Logistic Regression Model for a given dataset.	2
7	Develop Decision Tree Classification model for a given dataset and use it to classify a new sample.	2
8	Implement Naïve Bayes Classification in Python	2
9	Build KNN Classification model for a given dataset.	2
10	Write a python program to implement K-Means clustering Algorithm.	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Tom M. Mitchell, "Machine Learning", McGraw-Hill, 2010
2.	Bishop, Christopher. Neural Networks for Pattern Recognition. New York, NY: Oxford University Press, 1995.
3.	Introduction to Machine Learning, Fourth Edition By Ethem Alpaydin-2020-MIT Press

Reference Books	
1.	Ethem Alpaydin, (2004) "Introduction to Machine Learning (Adaptive Computation and Machine Learning)", The MIT Press.
2.	T. astie, R. Tibshirani, J. H. Friedman, "The Elements of Statistical Learning", Springer(2nd ed.), 2009

Useful Links	
1.	https://www.jnit.org/wp-content/uploads/2020/04/Machine-Learning-Lab-Manual.pdf
2.	https://deepakdvallur.weebly.com/machine-learning-laboratory.html
3.	https://mrcet.com/pdf/Lab%20Manuals/MACHINE%20LEARNING%20LAB.pdf

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